Mechanical Chaos: A Spring Pendulum as a Duffing Oscillator

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Chaos limits long-term prediction because small differences in a system's initial state can grow into substantially different motion. A disk coupled to two springs and carrying an off-center mass provides a mechanical realization of a Duffing-like oscillator. The balance of spring and gravitational torques determines whether its potential has one well or two; a motor drives the motion and a magnet damps it. Free releases indicated one minimum over the sampled angles. Driven recordings contained both irregular motion and nearly repeating orbits. In the longest recording, the angular spread measured at a fixed motor phase decreased by a factor of 4.6 (conditional 95% resampling interval: 4.0–5.2), quantifying the transition toward repeatable motion. A separate time-series test was also consistent with a transition from chaotic to regular behavior. Noise and changes in motor speed prevent a unique attribution to deterministic chaos, but the increase in repeatability is well resolved.

1. INTRODUCTION AND THEORY

1.1 The Duffing oscillator and chaos

Deterministic motion need not remain predictable: a small uncertainty in the initial state of a chaotic system can grow until two trajectories differ by the full extent of the motion. A driven pendulum makes this loss of repeatability accessible in a simple mechanical system. Irregular motion may also be temporary, and a changing drive can obscure its origin [1,2]. This experiment uses free releases to identify the sampled potential and controlled driven recordings to characterize the response. The main question is how strongly the motion changes from irregular to repeatable, and whether a quantitative time-series test supports a chaotic interpretation.

The Duffing model describes a driven oscillator with a nonlinear restoring force,

$$\ddot{x} + \delta\dot{x} + \alpha x + \beta x^3 = \gamma\cos(\Omega t), \tag{1}$$

where x is a displacement, δ is the damping coefficient, γ is the drive strength, and Ω is its angular frequency. The coefficients α and β define the potential

$$V(x) = \frac{1}{2}\alpha x^2 + \frac{1}{4}\beta x^4. \tag{2}$$

For $\beta > 0$, positive α gives one minimum and negative α gives two (Fig. 1). The number of wells determines whether motion can switch between potential minima. A double-well oscillator can exhibit chaotic motion between wells for suitable forcing, but crossing the barrier alone does not imply chaos [3]. Chaos and transient irregular motion are also possible in a driven single-well system [1,4].

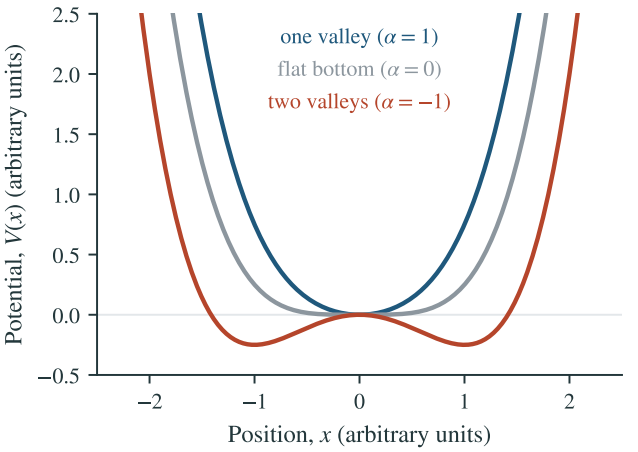


FIG. 1. The Duffing potential Eq. (2) for $\beta = 1$: one valley for $\alpha > 0$, a flat bottom for $\alpha = 0$, and two valleys for $\alpha < 0$.

1.2 From the pendulum to the Duffing equation

The apparatus consists of a disk carrying an off-center mass, coupled to two springs and driven by a motor that moves one spring end. A nearby magnet damps the motion. The compact torque diagram (Fig. 2) connects these components to the equation of motion. With angle θ measured from the position where the mass is directly above the axle,

$$I\ddot{\theta} + b\dot{\theta} + 2kr^2\theta - mgL\sin(\theta) = F_0\cos(\Omega t), \tag{3}$$

where I is the rotational inertia, b the damping coefficient, k each spring's stiffness, r the pulley radius, m the eccentric mass, L its distance from the axle, and g gravitational acceleration. For a drive arm of length A, the torque amplitude is $F_0 = krA$.

Using $\sin(\theta) \approx \theta - \frac{\theta^3}{6}$ and dividing by I gives Eq. (1) with $x = \theta$ and

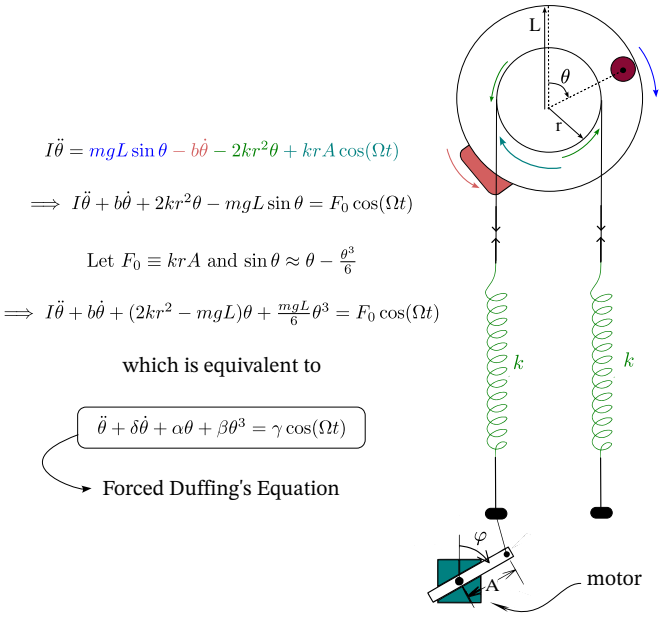


FIG. 2. Torque balance for the driven spring pendulum. The colored terms identify gravity (blue), magnetic damping (red), spring torque (green), and motor forcing (cyan). The angle θ is measured from the upright mass position; L is the distance from the axle to the mass, r the pulley radius, and A the drive-arm length. The small-angle expansion gives the Duffing form shown below the balance.

$$\delta = \frac{b}{I}, \quad \alpha = \frac{2kr^2 - mgL}{I}, \quad \beta = \frac{mgL}{6I}, \quad \gamma = \frac{F_0}{I}. \quad (4)$$

Near the upright position, spring torque is restoring while gravity is destabilizing; their balance sets the sign of α. The cubic approximation applies near this position and is not assumed valid for the measured swings of several radians. The potential shape is instead estimated directly from free releases.

The components and measurement connections are shown in Fig. 3. The encoder measures the disk angle, while the photogate detects the motor shaft at the same point in each revolution. These signals allow the pendulum state to be sampled at a fixed drive phase.

TABLE I. How each setting of the apparatus enters the Duffing equation Eq. (1) through Eq. (4). The added mass also changes I slightly.

Setting	Coefficient	Effect of increasing it
Motor voltage V	Ω	Faster drive
Drive arm A	$\gamma = krA/I$	Stronger drive
Magnet gap h	$\delta = b/I$	Weaker braking
Added mass m	α, β	Shallower well; two wells for $\alpha < 0$

1.3 Poincaré sections, bifurcation diagrams, and transient chaos

The number of variables needed to specify the state helps explain how chaos becomes possible. For a smooth autonomous system, whose equations have no explicit time dependence, bounded chaotic motion requires at least three state variables. An undriven pendulum has only angle and angular velocity; trajectories cannot cross in this plane because a given state has

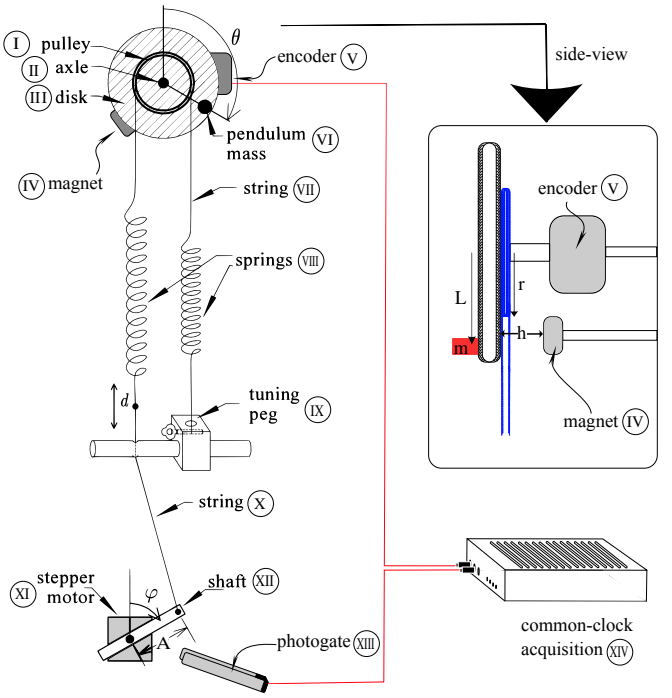


FIG. 3. The pendulum; the inset is a side view of the disk. The springs (VIII) pull through strings (VII) on the pulley (I) of radius r, turning the disk (III) on its axle (II). The mass (VI) at distance L supplies the gravitational torque, the magnet (IV) at gap h brakes the disk, and the tuning peg (IX) sets the spring tension. The motor (XI) turns the shaft (XII) through phase φ, and an arm of length A moves one spring end by d through the string (X). The encoder (V) records the angle θ, and the photogate (XIII) marks each pass of the shaft, both on one clock (XIV).

a unique future. The periodic drive adds its phase $\varphi = \Omega t$ as a third variable. The same angle and velocity can then lead to different subsequent motion if the motor is at a different phase. Nearby states can separate while the motion remains bounded, allowing a nonrepeating trajectory in the three-dimensional state space. Three variables permit chaos; they do not guarantee it.

A phase portrait plots angular velocity against angle. A Poincaré section samples these quantities once per drive cycle at a fixed phase, allowing successive cycles to be compared directly. Period-one motion returns to one point in an ideal measurement; period-two motion alternates between two. Noise and finite resolution broaden these points into clusters. A broad section records reduced repeatability but does not alone establish deterministic chaos [5,6].

Fig. 4 provides a conceptual comparison from a numerical double-well Duffing model [7]. Its bifurcation diagram plots the coordinate sampled after transients have decayed as drive strength is varied; branch splitting illustrates period doubling. Its Poincaré section shows the states approached at one fixed drive strength, the attractor. The experiment used fixed settings within each recording, so these illustrations are not an experimental bifurcation diagram. The analysis instead measures the narrowing of a Poincaré section and tests the time series for a change from irregular to regular behavior. Irregular motion that eventually approaches a periodic orbit is consistent with transient chaos, provided other causes of irregularity can be excluded.

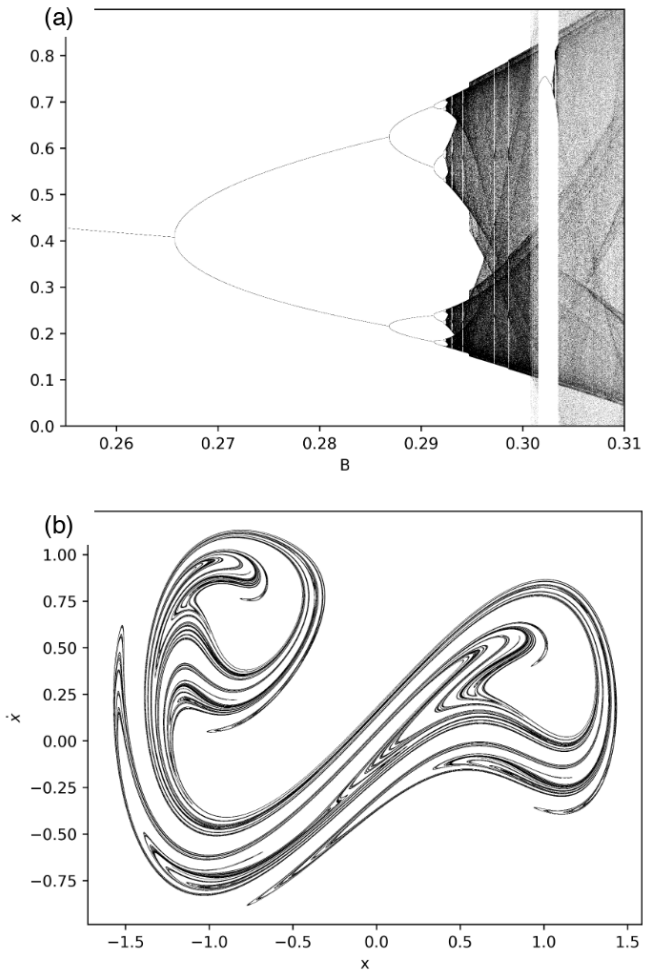


FIG. 4. Literature simulations of a double-well Duffing oscillator, adapted from Jani [7], not pendulum measurements. (a) Coordinate sampled once per drive cycle after the initial transient has decayed, plotted against forcing amplitude. Branch splitting illustrates period doubling. The source's x is a simulated coordinate, not the measured disk angle; its forcing amplitude B corresponds to γ in Eq. (1). (b) A Poincaré section in the simulated $(x, \dot{x})$ plane; the layered pattern illustrates a chaotic attractor. The coordinate scales and model parameters belong to that simulation. The plots provide conceptual context, not a calibrated prediction for the experimentally sampled single well.

2. METHODS

2.1 Instrument and acquisition

The PASCO chaotic pendulum (Fig. 3) was used for free releases and driven recordings [2,8]. The disk angle and angular velocity were recorded with a rotary encoder. Its smallest observed angle increment was 0.004 rad in driven runs and 0.008–0.017 rad in free releases. Photogate interruptions marked passages of the motor shaft; these event times remained usable although the instrument's timer-value field stayed at zero. The spring constants, lever arms and rotational inertia were not measured, so the Duffing coefficients could not be calculated from independent measurements of the apparatus.

The controlled series varied motor voltage, magnet gap or drive-arm length while holding the other settings fixed (Table II). No quantitative mass dependence was inferred because the mass assigned to each recording was uncertain. Runs with the same recorded settings were used to test reproducibility. String slips and changes in tension or arm position could change the actual conditions between runs; the size of these differences was not measured.

Duplicate time stamps were reduced to their first sample. For driven phase portraits, a 30 s running mean was subtracted from the angle to reduce slow drift of the angle zero; this does not recover the true motion during a sudden slip. The central 20%–85% of each moving record was used to reduce startup and stopping effects. Response frequency was taken from the largest spectral peak, with resolution $\frac{1}{T}$ for an analyzed interval of length T. Swing was defined as half the angle range between the 5th and 95th percentiles. Its variation within a run was estimated as half the absolute difference between the swings calculated from the first and second halves of the record.

Photogate events separated by less than 0.2 s were grouped as one shaft passage. Angle and angular velocity were interpolated to the first event in each group. The last 10 s of each recording were omitted from the motor-period test. The motor period is the time per revolution. The drive was classified as settled when this period remained within 2% of the median of its final 20 periods for at least 30 s. This test measures whether the drive has become steady. Pendulum periodicity was assessed separately from the Poincaré sections.

TABLE II. Settings for the three controlled series; one setting was varied while the other three were held at the values shown.

Series	Varied	Held fixed
Voltage	V = 5.52–8.02 V	A = 47.38 mm
		m = 17.5 g
		h = 4.12 mm
Gap	h = 6.83–18.48 mm	V = 5.52 V
		A = 47.38 mm
		m = 17.5 g
Drive arm	A = 52.44–62.74 mm	V = 5.52 V
		m = 17.5 g
		h = 13.64 mm

3. RESULTS AND ANALYSIS

3.1 The shape of the potential

Two free releases, F1 without added mass and F2 with 17.5 g added, were used to estimate the potential over the sampled angles (Fig. 5). They were treated as undriven because their waveforms decayed smoothly; the motor state was not logged. If little energy is lost during one swing, the total energy E is approximately constant and $\frac{\dot{\theta}^2}{2} + \frac{U(\theta)}{I} = \frac{E}{I}$, where U is the potential energy. Thus $-\frac{\dot{\theta}^2}{2}$ estimates the shape of $\frac{U(\theta)}{I}$ up to an additive constant. The first complete swing between upper turning points was used.

Both profiles have one resolved minimum over the sampled angular range. This supports motion in one accessible well, consistent with earlier observations on this apparatus [9,10]. Energy loss during the swing limits the reconstruction, and the

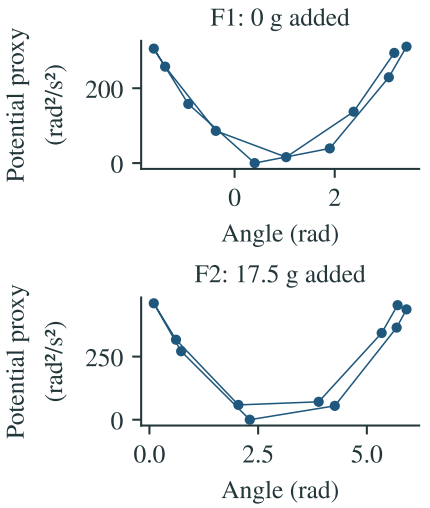


FIG. 5. Potential profiles estimated from free releases F1 (no added mass) and F2 (17.5 g added). Each profile plots $-\frac{\dot{\theta}^2}{2}$ from one complete swing, shifted to zero at its minimum. The vertical scale is energy per unit inertia, not energy in joules. One minimum is resolved in each sampled interval; the estimate assumes small energy loss during that swing.

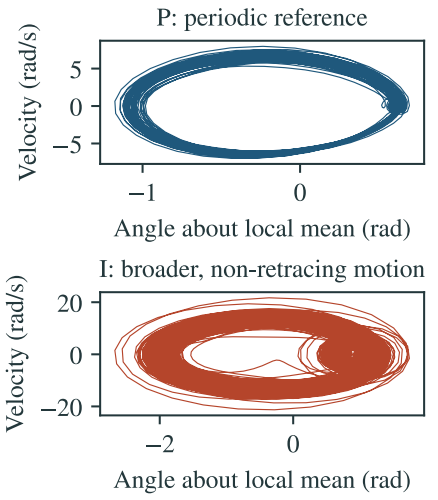


FIG. 6. Two selected phase portraits from the controlled series. P: 5.52 V, arm 47.38 mm, gap 4.12 mm. I: 5.52 V, arm 52.44 mm, gap 13.64 mm. Both used 17.5 g added mass. Angle is measured about its 30 s running mean over the central portion of each moving record. The broader portrait is accompanied by larger within-run variation, rather than a stable repeated orbit. The different gaps prevent attributing this pairwise contrast to arm length alone.

measurements do not exclude an unvisited minimum outside that range.

3.2 Changing one setting at a time

The long-arm series contained larger, less repeatable swings than the voltage series, whose portraits remained close to closed orbits. Two selected portraits (Fig. 6) illustrate that contrast: P is the lowest-voltage reference at the 47.38 mm arm; I is the shortest-arm record in the 52.44–62.74 mm series and also had the largest within-run swing variation. These examples illustrate distinct observed behaviors; a broad portrait alone does not quantify chaos.

The full controlled trends are summarized in Fig. 7. Response frequency increased with voltage at (0.2038 ± 0.0016) Hz/V, where the uncertainty is the least-squares fit standard error. Across ten logged records at 5.52 V, the frequency was (0.93 ± 0.03) Hz (sample standard deviation). The scatter at fixed voltage limits how reliably the voltage calibration predicts response frequency at different mechanical settings.

Over gaps of 6.83–18.48 mm, the swing ranged from 0.85 to 0.93 rad without a monotonic trend. At arm lengths of 52.44–62.74 mm it ranged from 1.5 to 2.0 rad, compared with about 0.9 rad in the 47.38 mm baseline series. The variation within a run is largest for the shortest arm in the long-arm series. The long-arm settings produced larger swings than the baseline, but the data do not establish a consistent increase with arm length. No quantitative mass dependence could be established.

3.3 Poincaré sections and settling times

The motor period met the settling criterion in 29 of 37 analyzable photogate records, after approximately zero to 2546 s. The remaining records ended without meeting this criterion; they do not establish that the pendulum would remain irregular indefinitely.

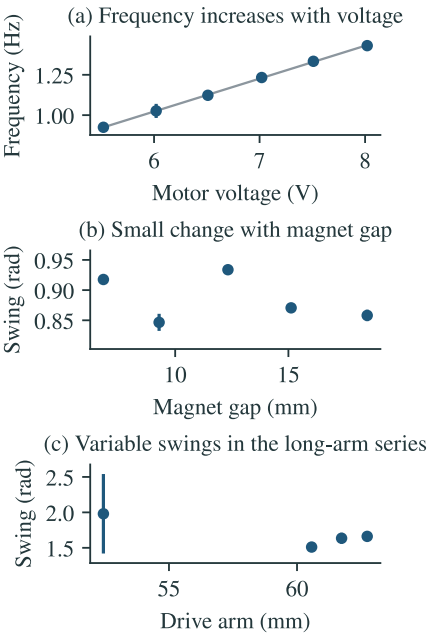


FIG. 7. Controlled response to voltage, magnet gap and drive-arm length, at the fixed settings in Table II. (a) Frequency and its linear fit; bars show spectral resolution $\frac{1}{T}$. (b,c) Swing, half the angle range between the 5th and 95th percentiles; bars show half the absolute difference between the swings in the first and second halves of each record. Separate vertical scales resolve the small variation with magnet gap and the larger variation with arm length. These bars describe within-run variation, not uncertainty from unlogged changes between recordings.

Runs with the same recorded settings nevertheless gave different transients. The selected pair S1 and S2 (Fig. 8) used the same logged voltage, arm, mass and gap, but their motor periods met the settling criterion at approximately 0 and 211 s. Late points in their Poincaré sections are more concentrated than earlier points. The different settling times demonstrate limited repeatability despite unchanged recorded settings, without identifying its cause.

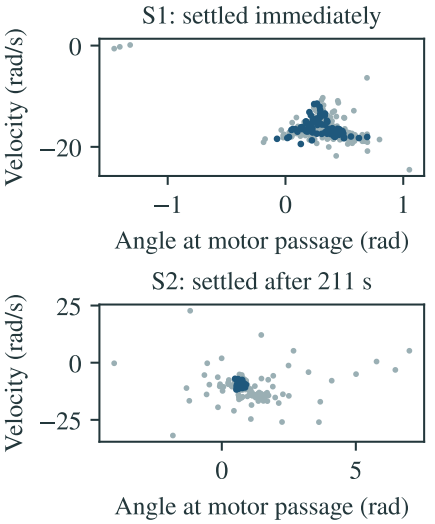


FIG. 8. Selected repeated-setting records S1 and S2 at 6.97 V, 72.66 mm arm, 14.5 g added mass and 19.82 mm magnet gap. Blue points are the final 60 s of the usable photogate record; pale points are earlier samples. Each point gives angle about a 30 s running mean and angular velocity at a motor passage. The motor periods met the settling criterion at different times despite the same recorded settings. Axis ranges retain all samples and differ between panels.

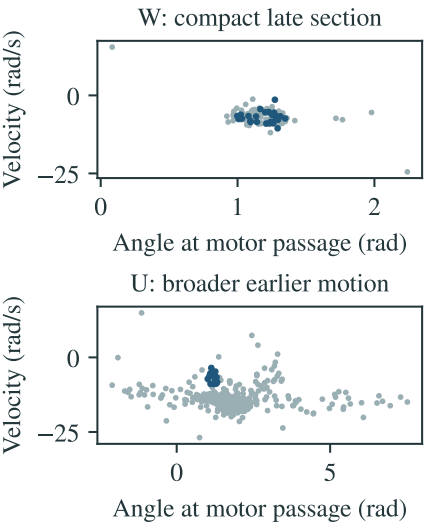


FIG. 9. A check on the drive-period criterion using W and U at 7.27 V, 80.93 mm arm, 15 g added mass and 4.28 mm magnet gap. Blue shows the final 60 s and pale points the earlier motion. Both motor periods continued to vary, although the late points formed relatively compact Poincaré clusters; U had substantially broader earlier motion. A failure to satisfy the motor-period criterion is not by itself a classification of the pendulum as chaotic.

Using the motor period to assess settling also has a limitation (Fig. 9). Neither W nor U met the motor-period criterion, although their late pendulum sections formed compact clusters. U sampled a much broader region earlier in the recording. A varying motor period therefore does not establish persistent pendulum chaos. The Poincaré section and motor-period record must be interpreted together.

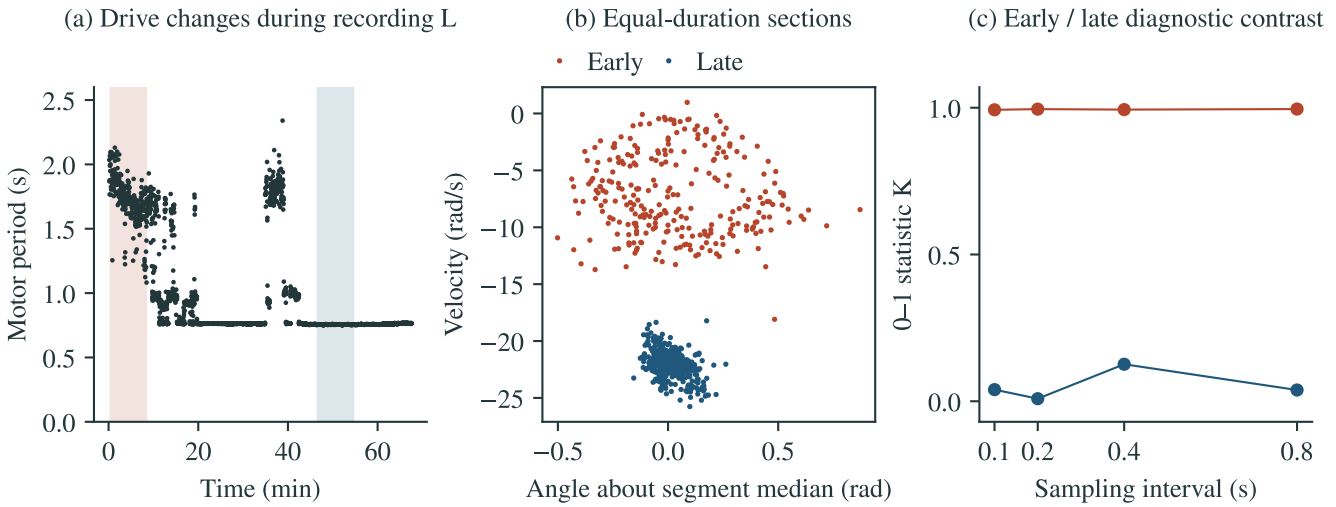


FIG. 10. Quantifying the transition in recording L. (a) Motor period; shaded intervals mark the equal-duration early (red, 30–500 s) and late (blue, 2800–3270 s) comparisons. The initial 4.98 s startup period lies outside the displayed vertical range; subsequent periods are retained. (b) Poincaré sections centered on their respective median angles. The angular spread decreases by a factor of 4.57 (conditional 95% block-resampling interval: 3.97–5.21). (c) The 0–1 statistic remains near one early and near zero late as the sampling interval changes. This supports the observed change in dynamics; noise and variable forcing prevent treating it as proof of deterministic chaos.

3.4 The hour-long run

The longest recording, L, resolves a substantial increase in repeatability within one run (Fig. 10). It lasted 4065 s at 7.27 V, an 80.93 mm arm, 15 g added mass and a 4.73 mm magnet gap. The early irregular interval, 30–500 s, was compared with 2800–3270 s in the late nearly periodic interval. Equal durations avoid confounding the comparison with record length. These intervals lie within the early and late regimes already identified from the record, rather than being selected by the diagnostic below.

The angular standard deviation at motor passage decreased from 0.262 rad (274 cycles) to 0.057 rad (622 cycles). Thus, the early section was $\frac{0.2616}{0.05724} = 4.57$ times wider: the pendulum returned much closer to the same angle on successive cycles late in the run. A circular block bootstrap, resampling consecutive groups of 16 cycles 4000 times, gave a conditional 95% interval of 3.97–5.21 for this ratio. Using 8 or 32 cycles per block changed the endpoints to within 3.87–5.25. These intervals estimate sampling variability conditional on stationarity within each window; they do not bound drift or photogate timing errors. The changing early drive limits that stationarity assumption. Subtracting a 30 s local mean changed the ratio from 4.57 to 4.64, leaving the contrast intact.

A second diagnostic was the correlation form of the 0–1 test [11]. For centered angle samples u_j at equal time intervals, the test forms a weighted sum and its mean squared displacement,

$$z_c(n) = \sum_{j=1}^{n} u_j \exp(ijc), \quad i^2 = -1,$$

$$D_c(n) = \left\langle |z_c(j+n) - z_c(j)|^2 \right\rangle_j .$$

Here n is a sample lag, c is a dimensionless phase increment, and the average is over all valid starting indices j. The statistic K is the median, over 100 values of c uniformly sampled between $\frac{\pi}{5}$ and $4\frac{\pi}{5}$, of the correlation between n and $D_c(n)$, using lags up to one tenth of the record. Bounded displacements give K near zero; approximately linear growth gives K near one. Under suitable deterministic conditions these correspond to regular and chaotic motion, respectively. K is not a growth rate or a probability of chaos.

At a 0.2 s sampling interval, K was 0.995 early and 0.009 late. Sampling intervals of 0.1–0.8 s gave 0.993–0.995 early and 0.009–0.126 late (Fig. 10, panel (c)). Splitting each interval into halves gave 0.983–0.990 early and 0.007–0.011 late. Changing the maximum displacement lag and the random phase increments also preserved the contrast. These are sensitivity ranges, not confidence intervals.

Periodic and chaotic logistic-map controls gave K near zero and one, respectively, but white noise also gave $K = 0.998$. Applying the test to once-per-cycle samples falsely assigned values near one to regular pendulum motion; it was therefore used only on the uniformly sampled angle series. Short records were also unreliable: a 50 s periodic reference approached one at the coarsest sampling interval, with only 63 samples. The reported comparison is restricted to the two 470 s windows of L, each containing at least 588 samples across the tested intervals.

3.5 Separation of repeated runs

Pairs of recordings with the same logged settings separated rapidly in recorded angle, as illustrated by D1–D3 (Fig. 11). These differences measure reproducibility between runs. A Lyapunov exponent would instead describe the exponential separation of nearby full states under the same dynamics [6]. The initial angle, angular velocity and motor phase were not jointly matched here, so the curves cannot directly provide that exponent.

Exploratory exponential fits were not stable under changes in mean subtraction, time alignment and fit interval; the inferred rates changed sign for two of the three pairs. The fits and rate values are therefore omitted from the figure and the reported results. The comparison retains the measured separation curves, without assigning them an exponential growth rate.

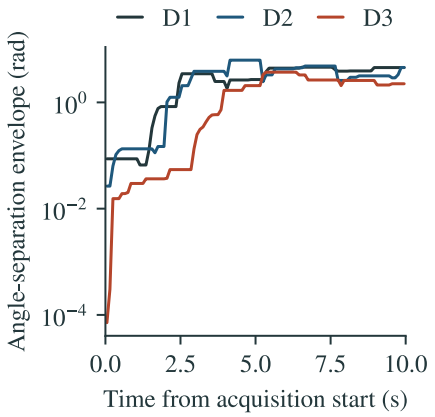


FIG. 11. First 10 s of recorded-angle separation for three pairs with the same logged settings. D1 compares S1 with a third repeat and D2 with S2 (Fig. 8); D3 compares U with W (Fig. 9). After subtraction of a centered 2 s running mean, absolute differences are summarized by a trailing 1 s maximum sampled every 0.1 s. The curves show rapid loss of agreement between recordings. The full initial states and forcing were not matched, so no Lyapunov exponent is assigned and no exponential fit is drawn.

4. DISCUSSION

The long recording establishes a measurable transition toward repeatable motion: the Poincaré angular spread decreases by a factor of 4.6 and the 0–1 statistic changes from near one to near zero. Agreement between these diagnostics strengthens the distinction between the two intervals. It does not uniquely establish deterministic chaos, since noise can also produce a high K [12].

The changing drive is a measured limitation. Between the compared intervals, the median motor period changed from 1.722 to 0.756 s, while its standard deviation relative to its mean decreased from 9.9% to 0.18%. Even within the early interval, the median period changed from 1.812 s in the first half to 1.664 s in the second. The two regimes therefore cannot be interpreted as transients under identical constant-frequency forcing. The resampling interval on section width quantifies finite-record variability, not this change in experimental conditions.

The observations are consistent with transient chaotic behavior in the coupled motor-pendulum apparatus. They establish increased repeatability more securely than they establish its cause. The 0–1 statistic survives the sampling and duration checks in L, but its response to noise and its failures on short or stroboscopic controls prevent a definitive chaos classification. No validated fractal dimension is available either: such an estimate would require stable scaling of the number of occupied phase-space cells as their size decreases, verified against a regular control [13].

The motor-period criterion remains useful for describing drive steadiness. Widening its tolerance from 2% to 3% changed no settled/unsettled classification and shifted settling times by at most 15 s. A steady motor period and periodic pendulum motion are nevertheless separate conditions, as the compact sections with a varying drive demonstrate (Fig. 9).

5. CONCLUSION

The spring pendulum exhibited one sampled potential minimum and both irregular and nearly periodic driven motion. In the longest recording, the angular spread at a fixed motor phase narrowed by a factor of 4.6 (conditional 95% resampling interval: 4.0–5.2). The accompanying change in the 0–1 statistic supports a transition toward regular motion. This is a quantitative result on repeatability; noise and the changing drive prevent its unique identification as transient deterministic chaos. No validated Lyapunov exponent is reported.

A stronger chaos test would regulate the motor speed, record its phase, and acquire longer intervals under steady conditions. Repeating runs with matched initial angle, angular velocity and drive phase, together with an independently measured noise baseline, would allow trajectory separation and time-series diagnostics to be tested under matched forcing. Measuring the spring constants, lever arms and inertia would also permit an independent comparison with the Duffing model.

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